

Fig. 6.1 shows four generations of a family where several individuals are affected by MELAS syndrome.

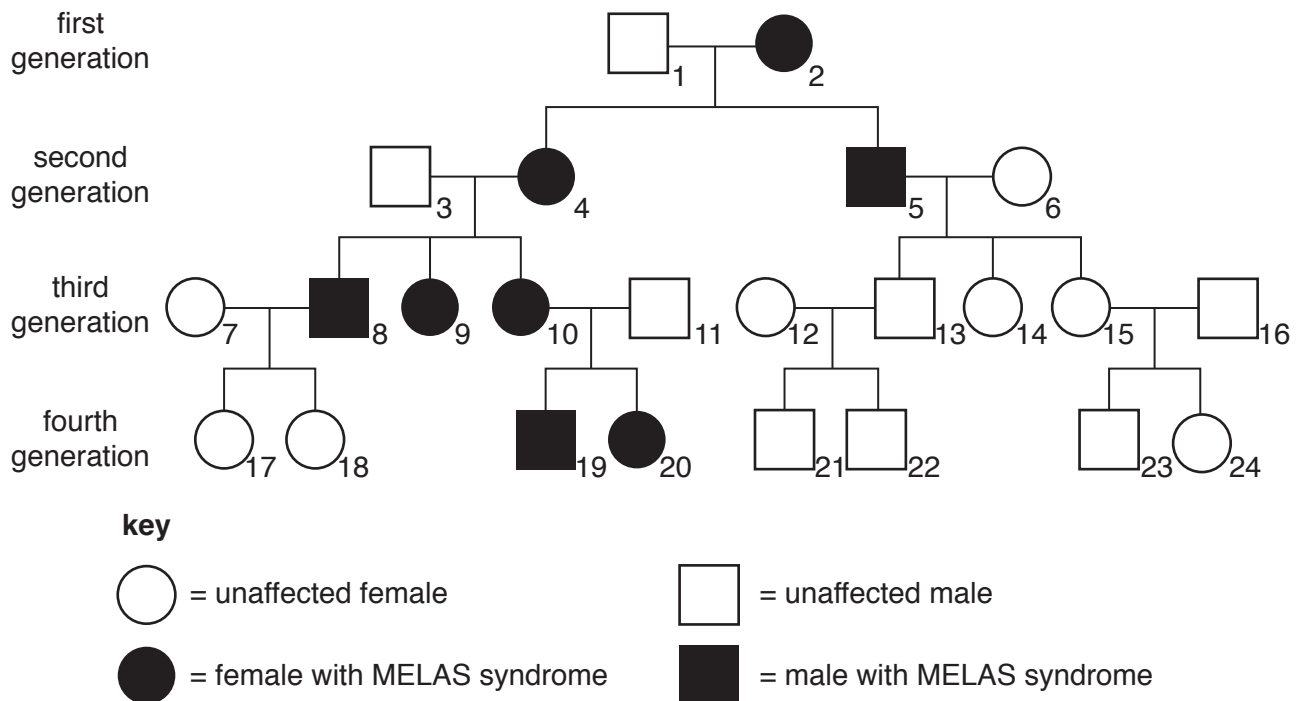


Fig. 6.1

- (a)** With reference to Fig. 6.1, state **and** explain the evidence that MELAS syndrome is an mtDNA disease and **not** a disease caused by a mutation in a gene on the X chromosome.

[4]

(b) Analysis of mtDNA can show how recently species have evolved from each other.

Describe the properties of mtDNA that make it suitable for the study of evolution.

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[Total: 7]